Vorlesung
Informationsvisualisierung

Kapitel 10
EINIGE STUDIENERGEBNISSE



Lecture
Information Visualization

Chapter 10
Some Empirical Results

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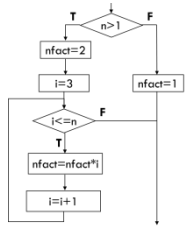
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Meta Study on Algorithm Animation (by Hundhausen et. al.)



- More than 40% of the 24 studies did not find any statistically significant results.
- „Thus, according to our analysis, *how* students use AV technology, rather than *what* students see, appears to have the greatest impact on educational effectiveness.“
- Several studies reported, that electronic learning material (multimedia or hypermedia) was more effective than lectures. The comparison with textbooks lead to less clear results.
- The way algorithm animations were used in learning experiments had more impact on the results than the actual quality of the visualization.
- In particular, algorithm animations were effective, if the learners had to (actively) solve
  - prediction tasks or
  - programming tasks.

Visual Programming and Flowcharts



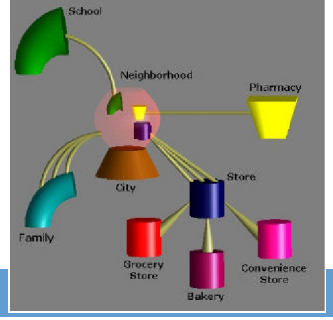
- In a quantitative study of a visual programming language Green and Petre [GP92] found, that in most cases the **diagrams were more difficult** to understand than the textual representation. This was true both for occasional as well as experienced programmers.
- Similarly, in an early study in 1977 [SMMH77], Flowcharts were more difficult to understand than the source code. Ten years later the experiment was **replicated** with additonal variables (measured values) and with **opposite outcome** [Sca89].

Visual Debugging



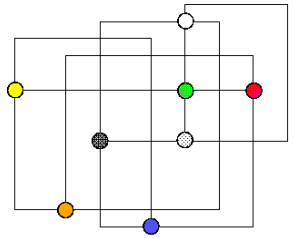
- Jones et al. evaluated Tarantula not with test persons, but with different versions of the same program each with a different (automatically injected) bug. They assessed how many of the buggy and the correct lines of code were shown in a reddish color.
- They used 1000 test cases and 60 versions of a program with 9500 lines of code. 20 versions contained only one (known) bug, 10 versions 2 bugs, 10 versions 3 bugs, 10 versions 4 bugs and 10 versions 5 bugs.
- They found, that **20% of the bug-free statements were drawn in a reddish color** (upper fifth of the color spectrum), whereas **most of the buggy statements were reddish colored**. Though, the number of reddish statements decreased with the number of bugs.

Geon and Class Diagrams



- Irani and Ware performed several experiments to compare Geon and UML-Class Diagrams.
- In one experiment [IW00] they showed test persons UML or Geon diagrams for 15 seconds. Then they showed them **diagrams of sub-sturctures** and asked, whether these had been part of the original diagram. Test persons answered by pressing the Y or N key. The average response was 4.3 seconds for Geon diagrams and 7.1 seconds for UML diagrams. Moreover, **the error rate was only 13% for Geon diagrams, but 26% for UML diagrams**.
- In another experiment [IW04] the test persons got **textual desciption** of a problem and 4 UML or Geon diagrams. They had to decide, which diagram was designed to solve the problem. The average **error rate was 15% for Geon diagrams and 36% for UML diagrams.**

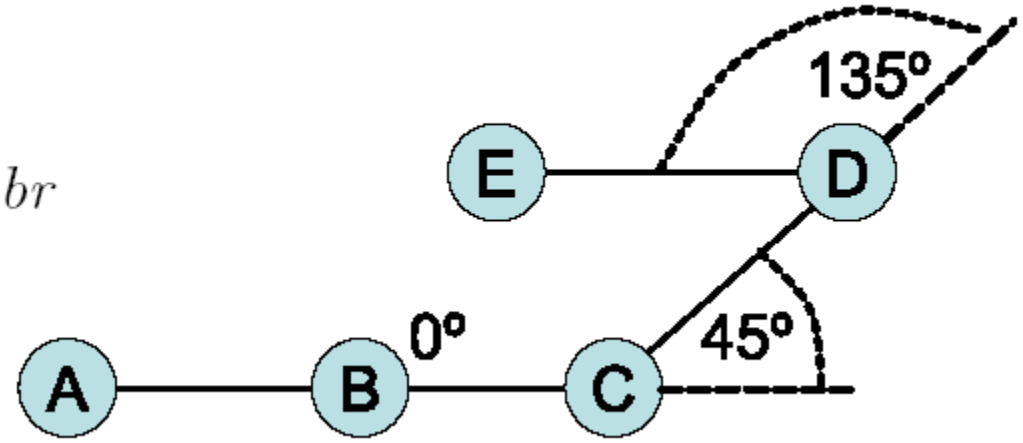
Graph Drawing Aesthetics



- Helen Purchase published many empirical studies investigating aesthetic criteria for graph layout.
  - For example, she found that the **number of edge crossings** is one of the most important criteria of the usual optimization goals used for graph drawing [PCJ96, Pur97].
  - Later she found that **continuity** (sum of orientation changes along a path) is an important factor as well (at least for the task of finding shortest paths) [WPCM02].

$$rt = 0.414\, spl + 0.406\, con + 0.317\, cr + 0.172\, br$$

rt: response time
spl: shortest path length
con: continuity
cr: number of edge crossings
br: number of branches



Source: Ware, C., Purchase, H., Colpoys, L., & McGill, M. (2002). Cognitive Measurements of Graph Aesthetics. Information Visualization, 1(2), 103–110. https://doi.org/10.1057/palgrave.ivs.9500013

Further reading or watching

- **39 studies about human perception in 30 minutes**
  - Kennedy Elliott
- https://medium.com/@kennelliott/39-studies-about-human-perception-in-30-minutes-4728f9e31a73#.m74fkz61j

- Video:

https://youtu.be/s0J6EDvlN30

